

Yuxuan Liu

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Education

University of Virginia

Start from 09/2025

Ph.D. Candidate in Computer Engineering

Tsinghua University

09/2021 – 07/2024

Master of Science in Data Science and Information Technology | GPA: **3.98**/4.00; Ranking: **8**/134

Core Courses: Advanced Signal Processing: Methods and Practice (A+), Nanogenerators and Self-powered Systems (A-), Operations Research (A-), Fundamentals of Information Theory (A), Course Project of Embedded Systems (A)

Beijing Jiaotong University

09/2017 - 07/2021

Bachelor of Engineering in Communications Engineering | GPA: **3.86**/4.00; Ranking: **7**/254

Core Courses: Signals and Systems (97/100), Digital Signal Processing (94/100), Digital Electronics Technology (96/100), Fundamentals of Wireless Communications (90/100), Principles of Communication Systems (92/100), Electromagnetic Field and Magnetic Waves (94/100)

Publications

- [J4] **Yuxuan Liu***, Nan Zhou*, Haoyang Wang, Fanhang Man, Jingao Xu, Fan Dang, Chaopeng Hong, Yunhao Liu, Xiao-Ping Zhang, Xinlei Chen, “CatUA: Catalyzing Urban Air Quality Insights through Mobile Crowd-Sensing”, **IEEE TMC**, 2025 (CCF-A, TH-CPL-A, SCI-Q1, [DOI](#))
- [J3] Rongye Shi*, Ji Luo*, Nan Zhou, **Yuxuan Liu**, Chaopeng Hong, Xiao-ping Zhang, Xinlei Chen, “Phy-APMR: A physics-informed air pollution map reconstruction approach with mobile crowd-sensing for fine-grained measurement”, **Elsevier Building and Environment**, 2025 (SCI-Q1, [DOI](#))
- [C9] **Yuxuan Liu***, Haoyang Wang*, Fanhang Man, Jingao Xu, Fan Dang, Yunhao Liu, Xiao-Ping Zhang, Xinlei Chen, “MobiAir: Unleashing Sensor Mobility for City-scale and Fine-grained Air-Quality Monitoring with AirBERT”, **ACM MobiSys’24** (Acceptance Rate≈16.3%, [DOI](#))
- [C8] Weichen Zhang, **Yuxuan Liu**, Xuzhe Wang, Xuecheng Chen, Chen Gao, Xinlei Chen, “Demo Abstract: Embodied Aerial Agent for City-level Visual Language Navigation Using Large Language Model”, **ACM/IEEE IPSN’24** ([DOI](#))
- [C7] Haoyang Wang, Xinyu Luo, Ciyu Ruan, Xuecheng Chen, Wenhua Ding, **Yuxuan Liu**, Xinlei Chen, “Poster: Fusing Event and Depth Sensing for Dynamic Objects Localization and Tracking”, **ACM HotMobile’24** ([DOI](#))
- [C6] Yuhan Cheng, Xuecheng Chen, Yixuan Yang, Haoyang Wang, **Yuxuan Liu**, Xinlei Chen, “Poster: Olfactory Sensing in Turbulent Airflow via Collaborative Robots”, **ACM HotMobile’24** ([DOI](#))
- [C5] Chenyu Zhao, Ciyu Ruan, Shengbo Wang, Jirong Zha, Haoyang Wang, Jiaqi Li, **Yuxuan Liu**, Xuzhe Wang, Xinlei Chen, “Demo Abstract: Bio-inspired Tactile Sensing for MAV Landing with Extreme Low-cost Sensors”, **ACM/IEEE IPSN’24** ([DOI](#))
- [J2] Xuecheng Chen, Haoyang Wang, Yuhan Cheng, Haohao Fu, **Yuxuan Liu**, Fan Dang, Yunhao Liu, Jinqiang Cui, Xinlei Chen, “DDL: Empowering Delivery Drones with Large-scale Urban Sensing Capability”, **IEEE JSTSP**, 2024 ([DOI](#))
- [C4] **Yuxuan Liu***, Haoyang Wang*, Chenyu Zhao, Jiayou He, Wenbo Ding, Xinlei Chen, “CaliFormer: Leveraging Unlabeled Measurements to Calibrate Sensors with Self-supervised Learning”, **ACM UbiComp/ISWC’23** (CPD Workshop, [DOI](#))
- [C3] Haoyang Wang, Fanhang Man, Zihan Wang, **Yuxuan Liu**, Xinlei Chen, Wenbo Ding, “Poster Abstract: TENG-enabled Self-powered Human-machine Interfaces for the Metaverse”, **ACM/IEEE IPSN’23** ([DOI](#))
- [C2] **Yuxuan Liu***, Yifei Sun*, Ziteng Wang, Xiaoli Qu, Dezhi Zheng, Xinlei Chen, “C-RIDGE: Indoor CO2 Data Collection System for Large Venues Based on Prior Knowledge”, **ACM SenSys’22** (DATA Workshop, [DOI](#))
- [C1] **Yuxuan Liu**, Xinyu Liu, Fanhang Man, Chenye Wu, Xinlei Chen, “FAD: Fine-grained Air Pollution Data Enables Smart Living and Efficient Management”, **ACM SenSys’22** ([DOI](#))
- [J1] **Yuxuan Liu**, “Design of an Internet of Things (IoT)-based shipping monitoring platform and database system”, **The Journal of New Industrialization**, 2020,10(03):126-128+133 (ISSN 2095-6649)

Conference Presentation

- Paper Presentation: MobiAir, ACM MobiSys’24, Tokyo, Japan, June 2024
- Paper Presentation: CaliFormer, ACM UbiComp/ISWC’23, Cancún, Mexico, October 2023
- Paper Presentation: C-RIDGE, ACM SenSys’22, Virtual, November 2022
- Demo Presentation: FAD, ACM SenSys’22, Virtual, November 2022

Role: Project Leader | Adviser: Prof. Xinlei Chen

- ***CaliFormer: Leveraging Unlabeled Measurements to Calibrate Sensors with Self-supervised Learning***
 - Proposed CaliFormer, a sensor calibration model based on representation learning to extract temporal- and spatial-invariant knowledge shared among tasks to address the calibration issue in environmental data.
 - Pioneered the integration of self-supervised learning into sensor calibration tasks to address the challenge of limited labels in environmental data.
 - Introduced a Transformer-based model architecture and designed a Masked Language Modeling (MLM) pre-training task to train the calibration model effectively.
 - The MAE of data calibrated by CaliFormer can be reduced to **16.54 $\mu\text{g}/\text{m}^3$** , achieving a **30%** improvement compared to the SOTA methods. Furthermore, the model exhibited strong robustness under different labeling rates.
- ***MobiAir: Unleashing Sensor Mobility for City-scale and Fine-grained Air-Quality Monitoring with AirBERT***
 - Proposed MobiAir, the first accurate city-scale and fine-grained air quality monitoring system based on MCS.
 - Developed AirBERT to separate the spatially mixed gas measurements by extracting non-stationary and non-Markovian features and modeling the mutual influences among sensor readings.
 - Designed a knowledge-informed training strategy to enhance performance of MobiAir and reduce the demand for city-scale labeled data.
 - Implemented a modular sensing front-end with separate chambers and active airflow design, which aims to collect highly mobile air quality data under high sampling rates.
 - City-wide deployment and experimental results demonstrate MobiAir reduces sensing errors by **96.7%** with only **44.9ms** latency, outperforming the SOTA baseline by **39.5%**.
- ***FAD: Fine-grained Air Pollution Data Enables Smart Living and Efficient Management***
 - Proposed FAD, a multi-source environmental data-driven system, using fine-grained air pollution data to realize smart city.
 - Designed a low-cost and high-precision portable sensing device to collect high-spatial-resolution air pollution data. It can be deployed on highly mobile vehicles (e.g., buses, drones) to perform Mobile Crowdsensing (MCS) tasks.
 - Designed four services: Advice for healthy life, Route planning, Toxic gas leakages finding & warning, and High pollution-emission sources locating to improve residential living quality and official city management efficiency.
 - Collaborate with Environmental Thinking Technology Co.,Ltd to deploy over **300** devices in **17** cities and regions across China, with a total annual device operating time exceeding **1.5 million** hours and a data volume exceeding **2 billion** records. Experiments show that the data spatial resolution can be up to **45m**, representing an improvement of over **40 times** compared to traditional static sensing methods.

Role: Project Leader | Adviser: Prof. Wenbo Ding

- ***TENG-enabled Self-powered Human-machine Interfaces for the Metaverse***
 - Proposed a human-machine interface (HMI) system based on the soft self-powered TENG (Triboelectric Nanogenerator) to address the limited degrees of freedom issue encountered in traditional HMIs.
 - Assist in implementing an ergonomically designed TENG sensor prototype deployed on a human joint to collect electrical signals generated during body movements (e.g., the arm), and utilizing simple Artificial Neural Networks (ANN) to reconstruct and visualize the angles of motion.
 - The recognition resolution of bending angle of human arms can be up to **15°**, and our system can map real movements in the digital world.
- ***Distributed Low-frequency Mechanical Energy Conversion based TENG Prototype***
 - Designed standalone layered TENG sensors with low-cost acrylic sheets, simulated wave motion, collected sensor data with electrostatic meter, and analyzed signal waveforms using LabVIEW.
 - Investigated the impact of materials, contact area, and movement speed on signal intensity by altering substrate materials, roller dimensions, and sensor tilt angles, and enhanced signal through stacking multiple layers of sensors.
 - Validated effectiveness of converting low-frequency mechanical energy into electrical energy by illuminating LEDs.

Role: Group Leader | Advised by Prof. Xinlei Chen & Prof. Dezhi Zheng

➤ **C-RIDGE: Indoor CO₂ Data Collection System for Large Venues Based on Prior Knowledge**

- Proposed C-RIDGE, a low-power sensing system for high-resolution CO₂ data collection in large venues.
- Proposed a power optimization and anomaly detection module based on prior physical knowledge in sensor design, and develop a sensor deployment strategy driven by data and physical models as the deployment method.
- Collaborate with BOCOG to deploy **45** C-RIDGE systems in the National Speed Skating Oval during the 2022 Beijing Winter Olympics, and carried out **15-day** data collection of CO₂ concentrations with data released on Zenodo.
- Experiments demonstrated that C-RIDGE achieved a **36.1%** reduction in power consumption. The anomaly detection module achieved a **100%** detection rate for anomalies and effectively identified abnormal trends.

Experience

➤ **Internship**

- Domain LLM Research & Development Intern @ Meituan, Beijing 05/2025-07/2025
- Research & Development Intern @ Environmental Thinking Technology Co.,Ltd, Shenzhen 01/2024-06/2024
- Research Assistant @ Shenzhen Institute of Artificial and Robotics for Society (AIRS) 09/2021-10/2022

➤ **Session Chair**

- 2023 ACM UbiComp/ISWC(CPD Workshop) 10/2023

➤ **Invited Reviewer**

- 2023 IEEE JSTSP 11/2023

Selected Honors

2024 | **Outstanding Thesis** (M.Sc.) of Tsinghua University & **Outstanding Graduate** of Beijing

ACM MobiSys Student Travel Grants

Dean's Scholarship, Tsinghua University

First place of the Best Poster, International Workshop on Learning and Information Theory

Third prize of TBSI Retreat Student Poster Contest, Tsinghua University

2023 | **The Award of BEST PRESENTATION in CPD Session**, ACM UbiComp/ISWC

Second-class Overall Excellence Scholarship, Tsinghua University

2017-2021 | **First-class Academic Excellence Scholarship**, Beijing Jiaotong University (top 3%)

First-class Arts Excellence Scholarship, Beijing Jiaotong University (top 5%)

Second-class Sports Excellence Scholarship, Beijing Jiaotong University (top 5%)

Merit Students, Beijing Jiaotong University (top 3%)

Honorable Mention of Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling (MCM/ICM)

Third prize at the College Student Computer Design Contest

Skills

Programming Language: Python, Matlab, C, Assembly Language, Latex, MySQL, Java, HTML, CSS, JSP

Software & Platform: VS Code, Pycharm, Matlab, Eclipse, Multisim, Tina-TI, Protues, Keil5, Wireshark, NS2

Extracurricular Activities

Director of brass section, Symphony Orchestra, Beijing Jiaotong University 2018 - 2020

- Organized 4 concerts and exhibitions, including "Call It a Concert" and "Music Salon" Themed Music Concert, etc.
- In charge of repertoires selection, participants coordination, weekly rehearsal organization.

Study monitor, School of Electronic and Information Engineering, Beijing Jiaotong University 2018 - 2021

- Engaged in weekly academic atmosphere construction and academic assistance activities.
- Organized study groups to prepare learning materials regarding the key knowledge for underprivileged individuals.
- Adjusted tutoring methods actively to adapt to students with diverse backgrounds and basis.

Lab assistant, Tsinghua-Berkeley Shenzhen Institute, Tsinghua University 2021 - 2024

- Assisted in organizing annual international retreat conference and student poster contest, which enables diverse international students to share their research equally and learn innovative approaches to solving problems from each other's perspectives.
- Hosted annual lab symposium which invites professors from top universities around the world to exchange research ideas and methods with students.
- In charge of daily lab tour, including introducing the lab's capabilities, ongoing projects, and research achievements to visitors in an attractive and accurate manner.
- In charge of bi-weekly academic seminars, including the arrangement of proceedings, periodic lectures to enhance speakers' presentation skills, and training for lab newcomers in paper selection and the framework of slide-making.